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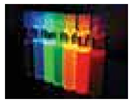

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# MEMBRANELESS ORGANELLES

## Probing liquid cellular structures

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Within the cell are a number of membraneless organelles (MLOs), without any structural boundary. These are composed of protein filaments and RNA. Some examples of MLOs include the ribosome and the cytoskeleton. Ribosomes manufacture proteins based on blueprints provided by RNA strands. They are like tiny little factories. Cytoskeleton are like the transport system of the cell, with bunches of protein strands with varying lengths. They transport the proteins to desired locations in the cell, so that the design encoded in the DNA can be stored. These organelles are drops of liquid, that exist within the cytoplasm of the cell, which is also a liquid. Due to the restrictions of imaging technologies, these membraneless organelles have been exceedingly difficult to study. The exact manner of their formation or the mechanisms of their function have not been easy to uncover. This is because high resolution imaging techniques, such as electron microscopy, have historically depended on the samples being suspended in vacuum. But living cells do not exist in a vacuum, and the samples undergo fundamental changes in such environments, which makes studying them useless. However, some new imaging techniques are providing insights about MLOs, and studying these structures is an emerging field.

### FORMATION

To probe MLOs, researchers first have to develop tools required to study them. The tools can subsequently have a much larger impact and uncover more details, beyond what the teams developing them initially find. Researchers from Princeton have created such tools, including one called Corelets, which uses genetically-engineered proteins that change their behavior and shapeshift when exposed to light. Another is CasDrop, a CRISPR-Cas9 based tool that can probe the chromatin, a mixture of DNA, RNA and proteins inside the nucleus of cells. Here, Cas9, a protein is used as a platform, which causes other proteins to bind to it when exposed to light. Both of these are considered what are called optogenetic tools.

The property of MLOs to remain separate is known as phase separation. Researchers from Princeton used CasDrop to create a phase diagram for the MLOs. This is a quantitative description of the precise ambient conditions during which the organelles are formed. Roughly, it is analogous to a map of temperature and pressure